

Task 15 — Intelligence Layer Integration & Model Governance

Data Analyst · Sprint C - Intelligence Layer · Task Brief

DATA ANALYST

PlaceMux · Phase 3 · Task Brief

TASK 15 / 25

Enterprise Grade · Expert

Sprint C - Intelligence Layer · Mastery 83%

Focus

Establish model-performance monitoring in production and integrate the intelligence layer's metrics.

What to build / complete

- Live model performance dashboards (quality, drift, fairness, latency, cost)
- Alerting on model degradation and outcome shifts
- An integrated view linking model metrics to business metrics

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: The business can see whether the AI is actually helping - or quietly getting worse - at any moment.

Stage B — Build: Live model performance dashboards (quality, drift, fairness, latency, cost)

1. Define the metric, its source & the decision

Define precisely what “Live model performance dashboards (quality, drift, fairness, latency, cost)” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: Alerting on model degradation and outcome shifts

1. Define the metric, its source & the decision

Define precisely what “Alerting on model degradation and outcome shifts” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: An integrated view linking model metrics to business metrics

1. Define the metric, its source & the decision

Define precisely what “An integrated view linking model metrics to business metrics” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Show the live model dashboard and demonstrate an alert on a simulated degradation.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “Live model performance dashboards (quality, drift, fairness, latency, cost)” is complete, real, and demoable end-to-end.
- “Alerting on model degradation and outcome shifts” is complete, real, and demoable end-to-end.
- “An integrated view linking model metrics to business metrics” is complete, real, and demoable end-to-end.
- Good looks like: The business can see whether the AI is actually helping - or quietly getting worse - at any moment.

Worry if any of these are true

- Model monitored only on technical metrics.
- No alerting on degradation.
- Calling an experiment early because it looked good on day two.

Hand-off

- Model dashboards to AI-ML and leadership. Make sure the next person can pick it up without you.